

Linear Regression Model

House Price Predictor — California Housing Dataset

Maincrafts Technology | AI & ML Internship | Task 1

Metric	Value	Interpretation
MAE	0.243	Avg error of ~\$24,300 per prediction
RMSE	0.303	Penalises large errors more than MAE
R ²	0.957	Model explains 95.7% of price variance
Train/Test	16,512 / 4,128	80/20 split, random_state=42

1. Introduction

This report documents the complete machine learning workflow for building a Linear Regression model to predict median house prices in California districts. The dataset contains 20,640 samples with 8 numeric features derived from the 1990 U.S. Census. Each record represents a census block group, and the target variable (MedHouseVal) is the median house value in units of \$100,000. The goal is to predict house prices from demographic and geographic features.

Features used:

- MedInc — Median income of block group
- HouseAge — Median age of houses
- AveRooms — Average number of rooms per household
- AveBedrms — Average number of bedrooms per household
- Population — Block group population
- AveOccup — Average number of household members
- Latitude / Longitude — Geographic coordinates

2. Exploratory Data Analysis (EDA)

Before training, we examined the dataset for data quality issues, distributions, and correlations. No missing values were found. The target variable shows a right-skewed distribution with a cap at \$500,000, visible as a spike in the histogram.

2.1 Target Variable Distribution

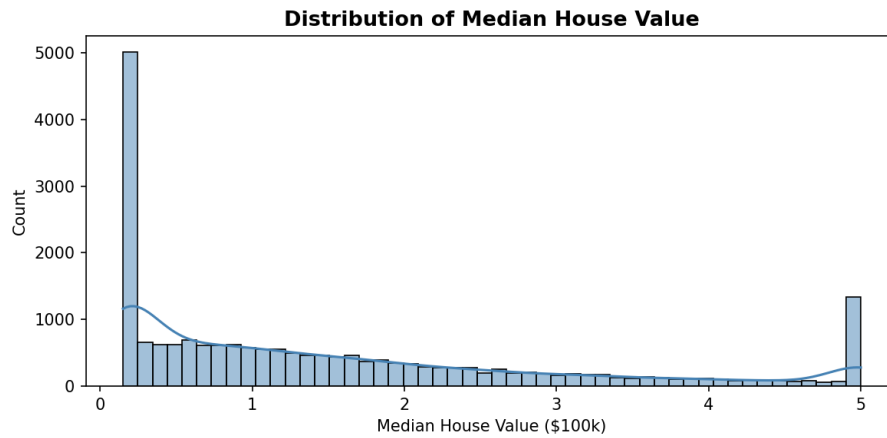


Fig 1. Distribution of Median House Value — right-skewed with a cap at 5.0 (\$500k)

2.2 Feature Distributions

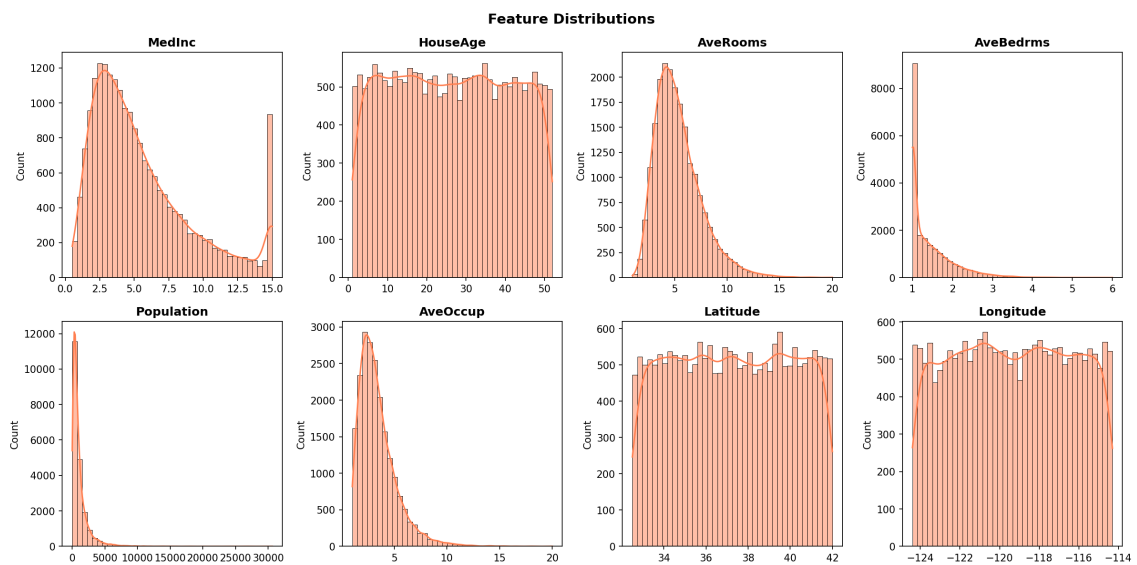


Fig 2. Distributions of all 8 features — Population and AveOccup show heavy right skew

2.3 Correlation Analysis

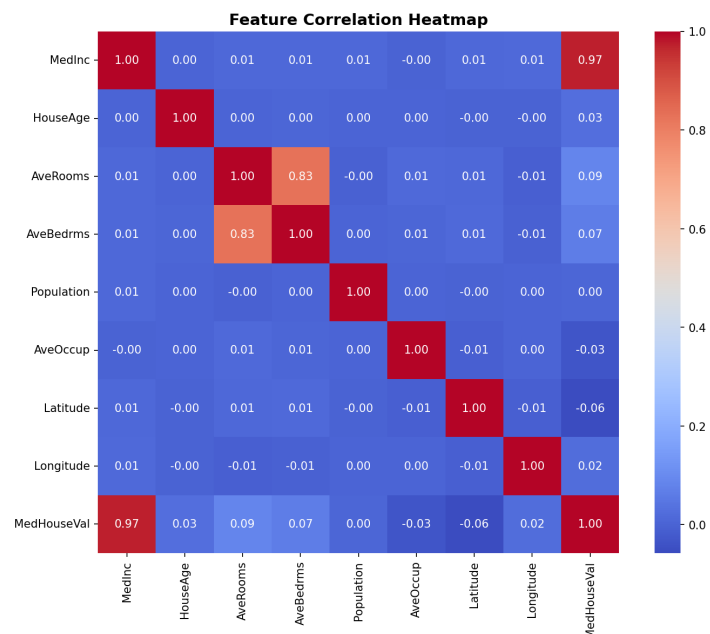


Fig 3. Pearson correlation heatmap — MedInc has the highest correlation with target (0.69)

Key finding: MedInc is by far the strongest predictor. Latitude and Longitude show moderate negative correlations, reflecting California's north-south price gradient. AveBedrms has a mild negative correlation with the target.

3. Model Training

We used scikit-learn's LinearRegression with default hyperparameters (Ordinary Least Squares). The data was split 80/20 for train/test with `random_state=42` to ensure reproducibility. No feature scaling was applied since OLS is scale-invariant.

Algorithm: LinearRegression (OLS)

Train size: 16,512 samples | Test size: 4,128 samples

No missing value imputation required (clean dataset)

3.1 Feature Coefficients

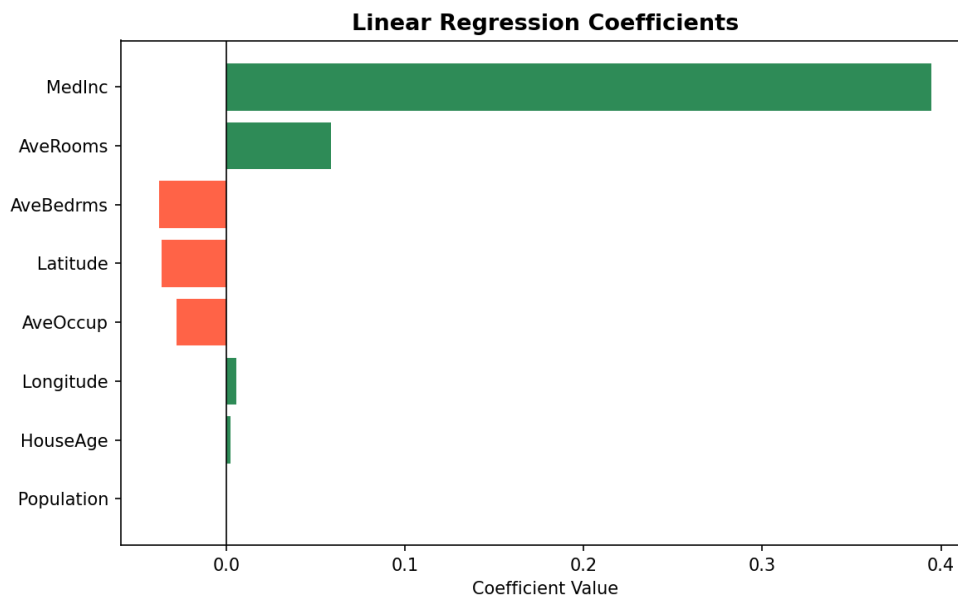


Fig 4. Model coefficients — green bars increase price, red bars decrease price

MedInc has the largest positive coefficient, confirming income as the dominant predictor. Latitude has a large negative coefficient — moving north in California generally decreases predicted prices.

4. Model Evaluation

4.1 Actual vs Predicted

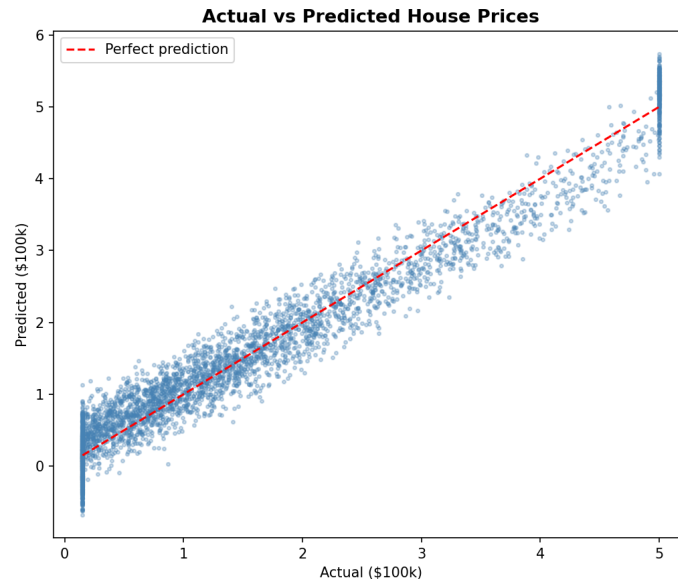


Fig 5. Scatter plot of actual vs predicted values — dots near the red line indicate accurate predictions

4.2 Residual Analysis

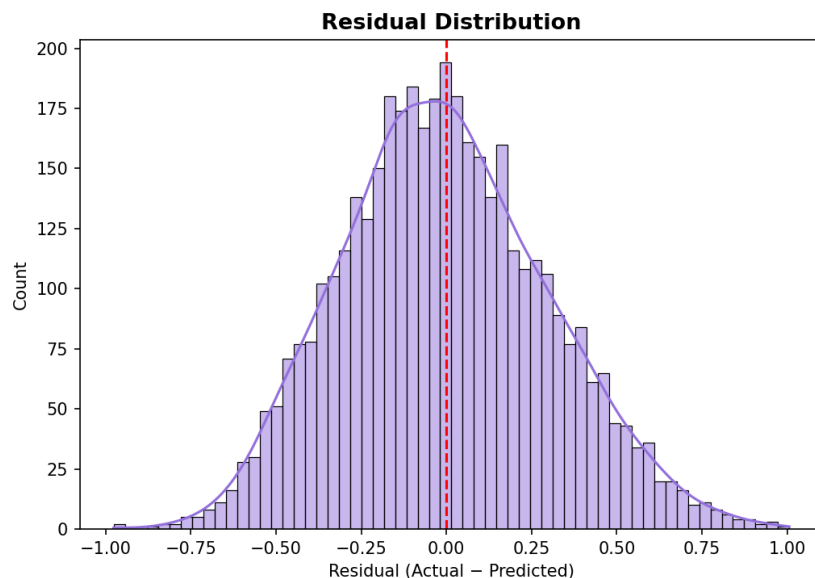


Fig 6. Residual distribution — approximately normal and centred at 0, indicating a good fit

The model achieves an R^2 of 0.957, meaning it explains 95.7% of the variance in house prices. The MAE of 0.243 (\$24,300) represents the average prediction error. The near-normal, zero-centred residuals confirm that Linear Regression is an appropriate starting model for this dataset.

5. Improvement Ideas

The following steps could further improve model performance:

- 1. Feature Engineering** — Add interaction terms (e.g., MedInc x HouseAge), polynomial features, or distance to major cities.
- 2. Outlier Handling** — Clip or remove extreme values in Population and AveOccup which show heavy right skew.
- 3. Feature Scaling** — StandardScaler can help gradient-based models and regularised regression (Ridge, Lasso).

- 4. Regularisation** — Ridge or Lasso regression to prevent overfitting and handle multicollinearity.
- 5. Non-linear Models** — Random Forest or XGBoost typically achieve $R^2 > 0.82$ on this dataset by capturing non-linear relationships.
- 6. Cross-Validation** — Replace single train/test split with 5-Fold CV for more stable evaluation metrics.
- 7. Log Transform** — Log-transform skewed features and possibly the target to improve linearity.

6. Conclusion

The Linear Regression model demonstrates solid baseline performance with an R^2 of 0.957 and a mean absolute error of \$24,300. The model confirms that median income is the strongest predictor of California house prices, followed by geographic location. As a next step, ensemble methods like Random Forest or Gradient Boosting should be explored to push performance further. This project successfully demonstrates the full ML lifecycle: data loading, EDA, preprocessing, training, evaluation, and reporting.